

Transport systems

SUBSTANCES ARE MOVED AROUND INSIDE LIVING THINGS IN A VARIETY OF WAYS.

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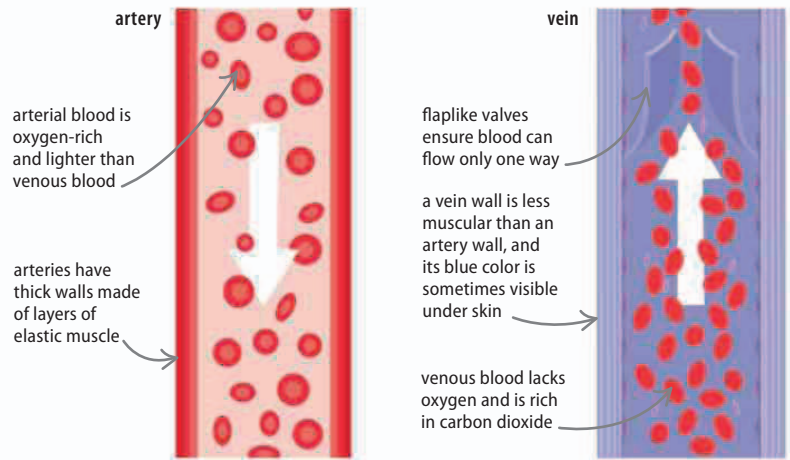
The cells in a multicellular organism are specialized into certain roles and cannot survive on their own. The body's transport system brings them what they need to stay alive, and takes away their waste materials.

Circulation

Animals transport substances around their bodies in a liquid. In vertebrates, this liquid is blood, pumped along by a heart (or hearts) through a series of pipes, or vessels. Blood vessels reach all parts of the body, narrowing to thin-walled capillaries that deliver materials to cells by diffusion.

▷ Arteries and veins

The vessels that carry blood away from the heart are called arteries. They pulsate to push blood along, which can be felt through the skin in some places. Veins bring blood back to the heart.

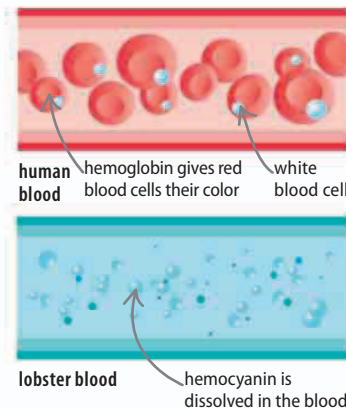


Composition of blood

Blood contains hundreds of compounds. About 55 percent of blood is a watery mixture known as plasma. This contains dissolved ions, hormones, and several proteins, such as the ones that form blood clots and scabs to seal breaks in vessels. The rest of the blood is made up of red and white blood cells and platelets.

Blood color ▷

Blood looks red because most of its cells contain an iron-rich pigment called hemoglobin. This substance bonds with oxygen arriving via the lungs and delivers it to body cells. A few invertebrates use copper-rich hemocyanin to do this, which makes their blood blue.



plasma contains many substances dissolved in it, such as carbon dioxide, which is produced as waste by cells

one in 20 blood cells are white blood cells, which defend the body against disease

oxygen-carrying red blood cells make up the majority of blood—there are five billion in every milliliter

▽ Red blood cells

Hemoglobin, the body's oxygen carrier, is held in red blood cells. These have a curved doughnut shape to maximize their surface area for collecting oxygen.

